

Nouns Conc-Abs: Graphs for Accuracy

author

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```
table(d$Task,d$Label)
```

```
##
##           test_conc test_val
## Concrete      4920      0
## Valence        0     4920
```

```
print(unique(d$Word))
```

```
## [1] "resentment" "chocolate" "sunshine" "grief" "garbage"
## [6] "failure" "gratitude" "hug" "loneliness" "fireplace"
## [11] "envy" "love" "rust" "smoke" "vomit"
## [16] "mold" "freedom" "filth" "rot" "kindness"
## [21] "rain" "laughter" "home" "peace" "flowers"
## [26] "mud" "guilt" "injury" "regret" "beach"
## [31] "stench" "shame" "compassion" "despair" "hope"
## [36] "joy" "inspiration" "happiness" "anxiety" "music"
```

```
table(d$Key_value_F)
```

```
##
## abstract concrete negative positive
##      2520      2400      2520      2400
```

```
print(unique(d$Word))
```

```
## [1] "resentment" "chocolate" "sunshine" "grief" "garbage"
## [6] "failure" "gratitude" "hug" "loneliness" "fireplace"
## [11] "envy" "love" "rust" "smoke" "vomit"
## [16] "mold" "freedom" "filth" "rot" "kindness"
## [21] "rain" "laughter" "home" "peace" "flowers"
## [26] "mud" "guilt" "injury" "regret" "beach"
## [31] "stench" "shame" "compassion" "despair" "hope"
## [36] "joy" "inspiration" "happiness" "anxiety" "music"
```

Summary Stats

```
agr <- d %>%
  group_by(Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            SD = sd(Accuracy))
print(agr)
```

```
## # A tibble: 2 x 3
##   Task      MeanAccuracy      SD
```

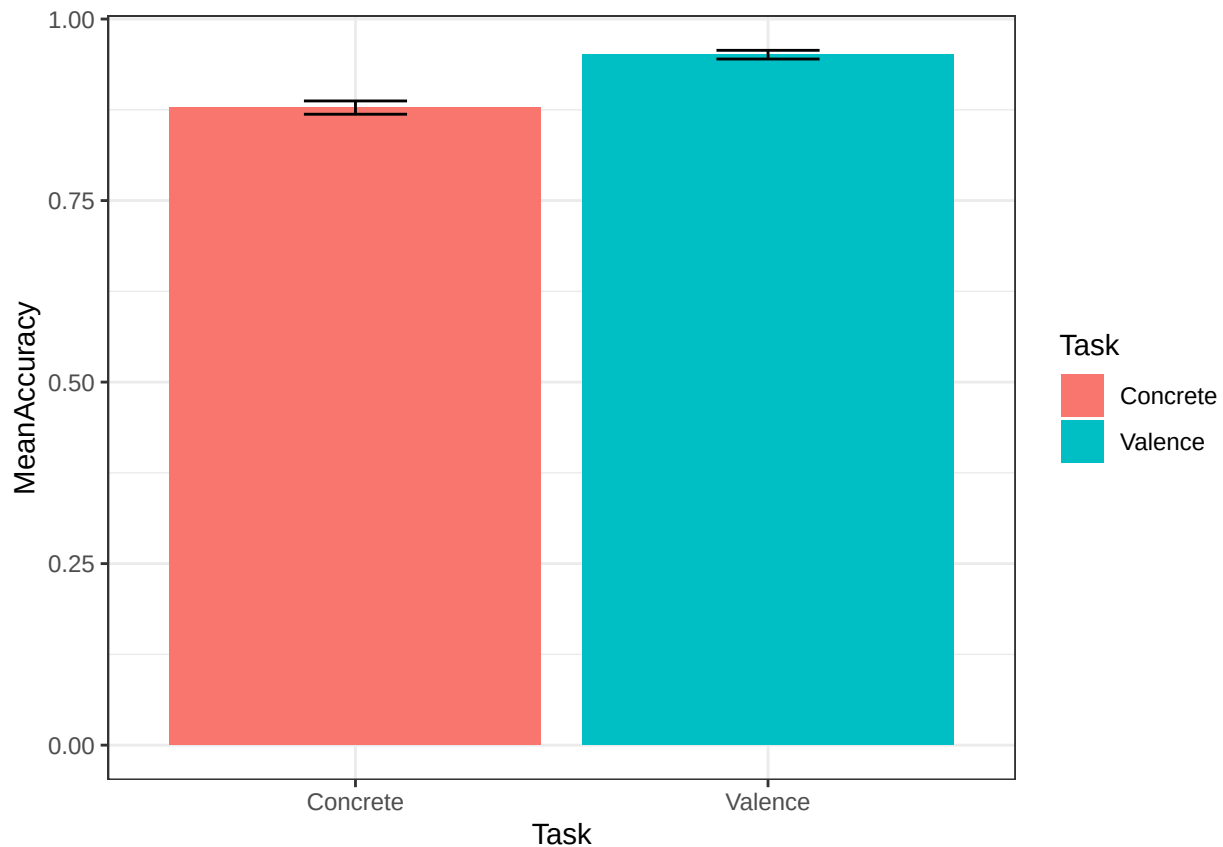
```
##   <chr>          <dbl> <dbl>
## 1 Concrete      0.878 0.327
## 2 Valence       0.951 0.216
```

Graph Accuracy by Word

Overall Accuracy

```
agr <- d %>%
  group_by(Task) %>%
  reframe(MeanAccuracy = mean(Accuracy),
          CILow = ci.low(Accuracy),
          CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
         YMax = MeanAccuracy + CIHigh)
# View(agr)

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Task,y=MeanAccuracy, fill=Task)) +
  geom_bar(position=dodge,stat="identity") +
  geom_errorbar(aes(ymin=YMin,ymax=YMax),width=.25,position=position_dodge(0.9))
```



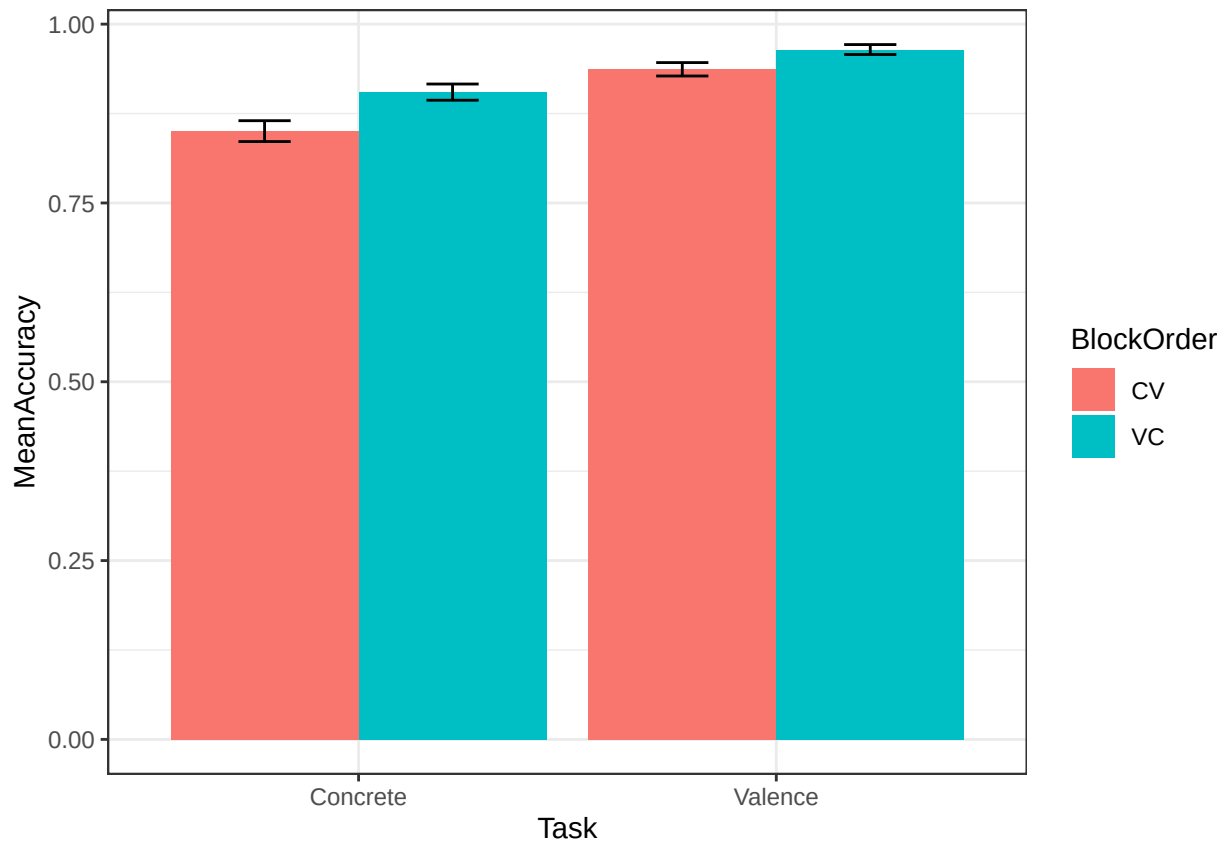
```
agr <- d %>%
  group_by(Task,BlockOrder) %>%
  reframe(MeanAccuracy = mean(Accuracy),
          CILow = ci.low(Accuracy),
```

```

    CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
         YMax = MeanAccuracy + CIHigh)
# View(agr)

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Task,y=MeanAccuracy, fill=BlockOrder)) +
  geom_bar(position=dodge,stat="identity") +
  geom_errorbar(aes(ymin=YMin,ymax=YMax),width=.25,position=position_dodge(0.9))

```



Mean Accuracy by Word / Task

Looking at only the first block

```

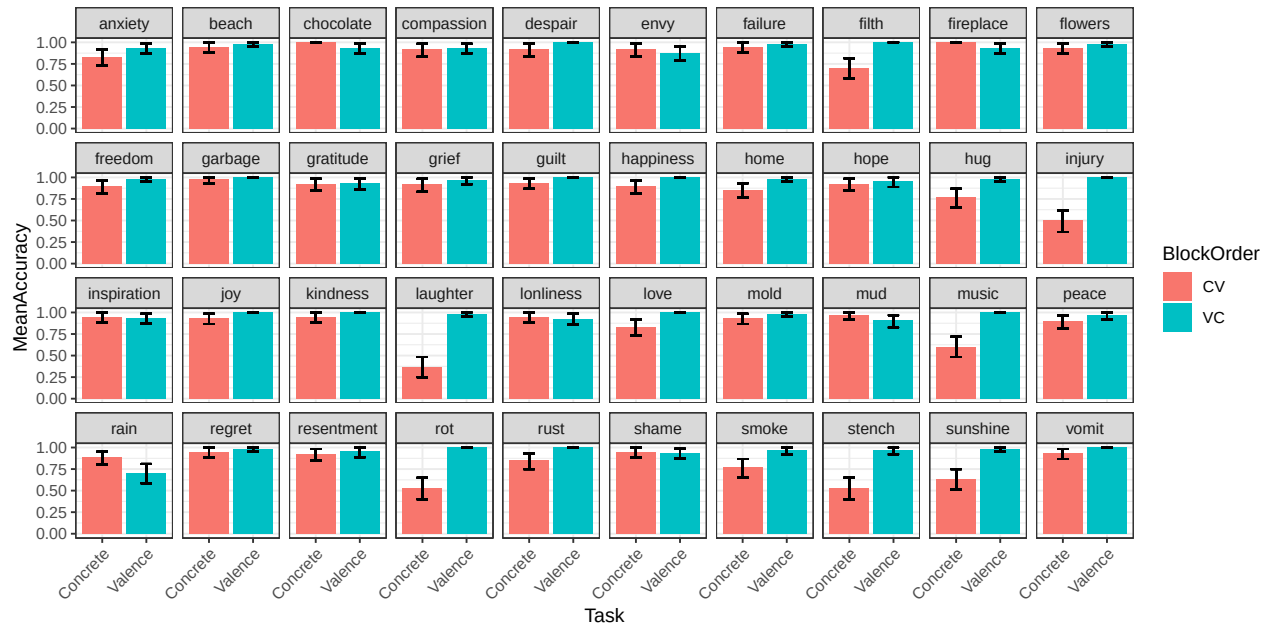
agr <- d %>%
  group_by(Task,Word,BlockOrder) %>%
  filter((Task == "Valence" & (BlockOrder == "VC") |
         (Task == "Concrete" & (BlockOrder == "CV"))) %>%
  mutate(MeanAccuracy = mean(Accuracy),
         CILow = ci.low(Accuracy),
         CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
         YMax = MeanAccuracy + CIHigh)

agrr <- agr %>%
  group_by(Word,Task) %>%
  select(Word,Task,MeanAccuracy) %>%

```

```
unique()
```

```
dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Task,y=MeanAccuracy,fill=BlockOrder)) +
  geom_bar(position=dodge,stat="identity") +
  facet_wrap(~Word,ncol=10) +
  geom_errorbar(aes(ymin=YMin,ymax=YMax),width=.25,position=position_dodge(0.9)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



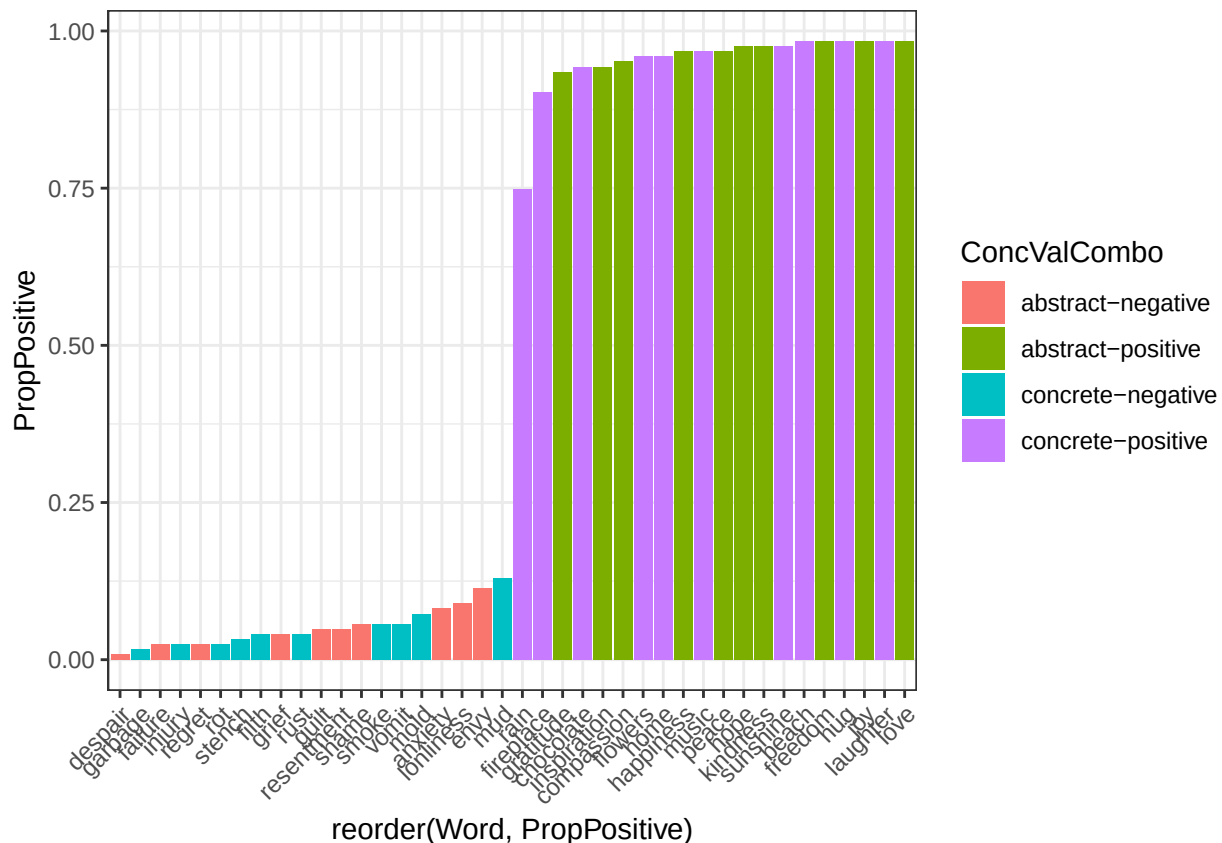
PropPositive and PropConcrete

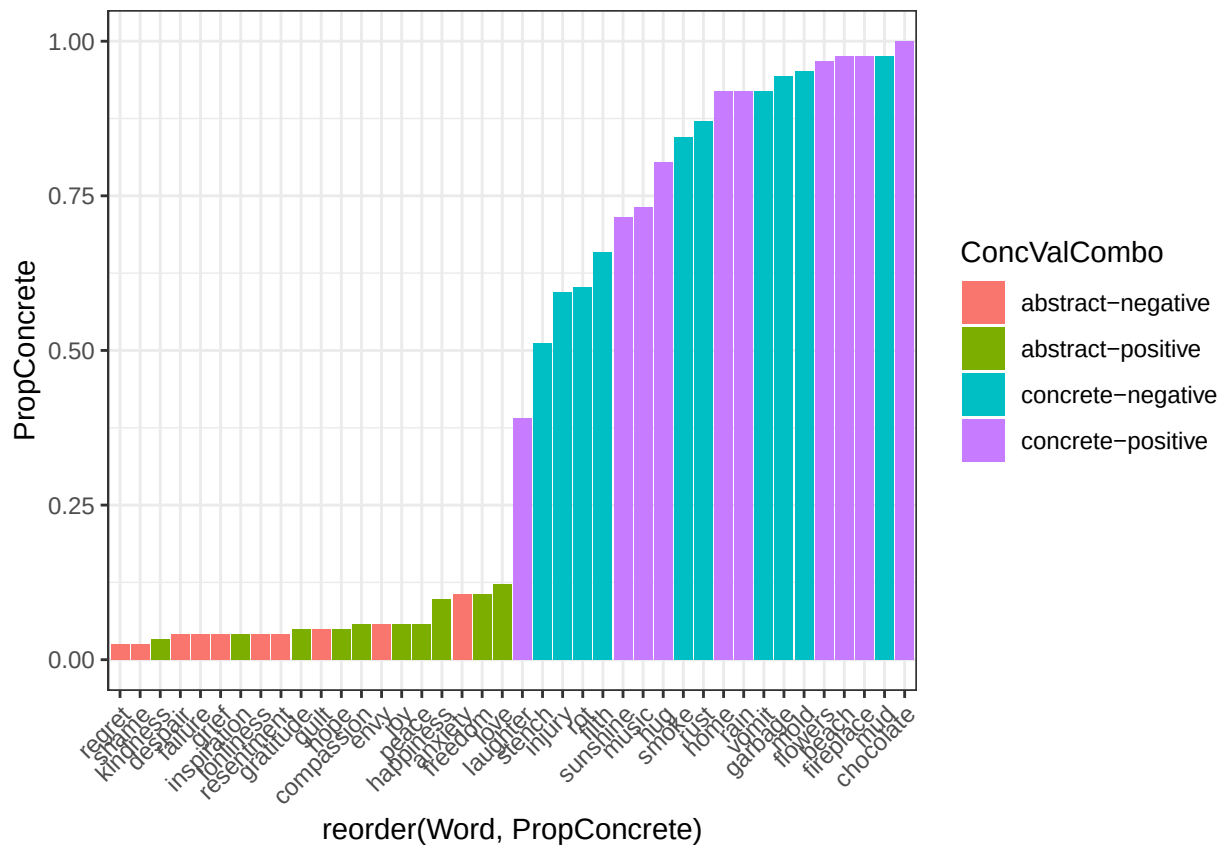
```
val <- d %>%
  filter(Task == "Valence") %>%
  # filter(Word %in% conc$Word) %>%
  group_by(Word, ConcValCombo) %>%
  mutate(Response.n = as.numeric(factor(Response, levels = c("negative", "positive")) - 1) %>% # Conv
  summarize(PropPositive = mean(Response.n))
```

```
## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.
```

```
# filter(PropPositive > .1 | PropPositive < .9)
```

```
dodge = position_dodge(.9)
ggplot(data=val, aes(x=reorder(Word,PropPositive),y=PropPositive,fill=ConcValCombo)) +
  geom_bar(position=dodge,stat="identity") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



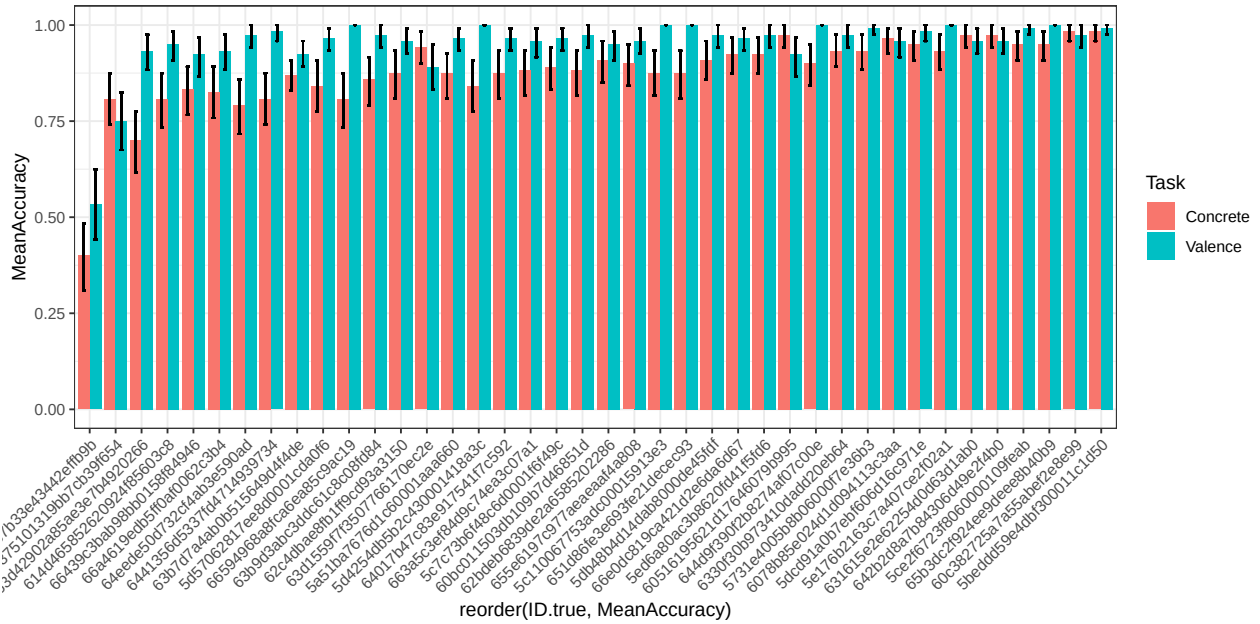


```
# guides(fill = "none")
```

Accuracy by Participant

```
agr <- d %>%
  # filter(PennElementType == "Selector") %>%
  select(ID.true, Task, Accuracy) %>%
  group_by(ID.true, Task) %>%
  mutate(MeanAccuracy = mean(Accuracy),
         CILow = ci.low(Accuracy),
         CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
         YMax = MeanAccuracy + CIHigh)

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=reorder(ID.true, MeanAccuracy), y=MeanAccuracy, fill=Task)) +
  geom_bar(position=dodge, stat="identity") +
  geom_errorbar(aes(ymin=YMin, ymax=YMax), width=.25, position=position_dodge(0.9)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



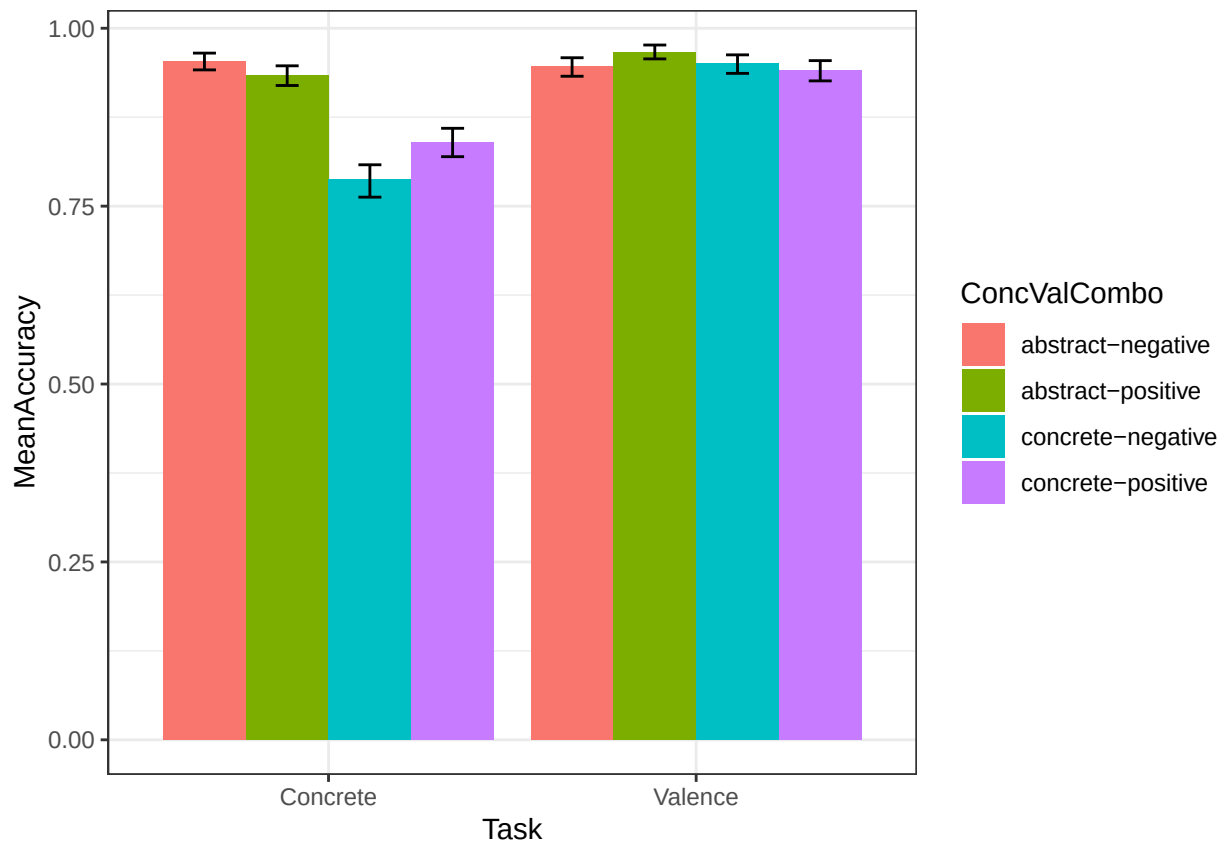
```
# guides(fill = "none")
```

Mean Accuracy by ConcValCombo

```
agr <- d %>%
  group_by(Task, ConcValCombo) %>%
  summarize(MeanAccuracy = mean(Accuracy), CILow = ci.low(Accuracy), CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow, YMax = MeanAccuracy + CIHigh)

## `summarise()` has grouped output by 'Task'. You can override using the
## `.groups` argument.

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Task, y=MeanAccuracy, fill=ConcValCombo)) +
  geom_bar(position=dodge, stat="identity") +
  # facet_wrap(~Task) +
  geom_errorbar(aes(ymin=YMin, ymax=YMax), width=.25, position=position_dodge(0.9))
```



```
# theme(axis.text.x = element_text(angle = 45, hjust = 1))
# guides(fill = "none")
```

Remove participants who aren't accurate

```
length(unique(d$ID.true))
```

```
## [1] 40
```

```
inacc.parts <- d %>%
  group_by(ID.true, Task) %>%
  summarise(MeanAccuracy = mean(Accuracy)) %>%
  filter(MeanAccuracy < .75)
```

```
## `summarise()` has grouped output by 'ID.true'. You can override using the
## `.groups` argument.
```

```
# How many participants have Accuracy < .75?
length(unique(inacc.parts$ID.true))
```

```
## [1] 2
```

```
d.inaccurate.removed <- d %>%
  anti_join(inacc.parts, by = "ID.true")

# Sanity check
length(unique(d.inaccurate.removed$ID.true))
```

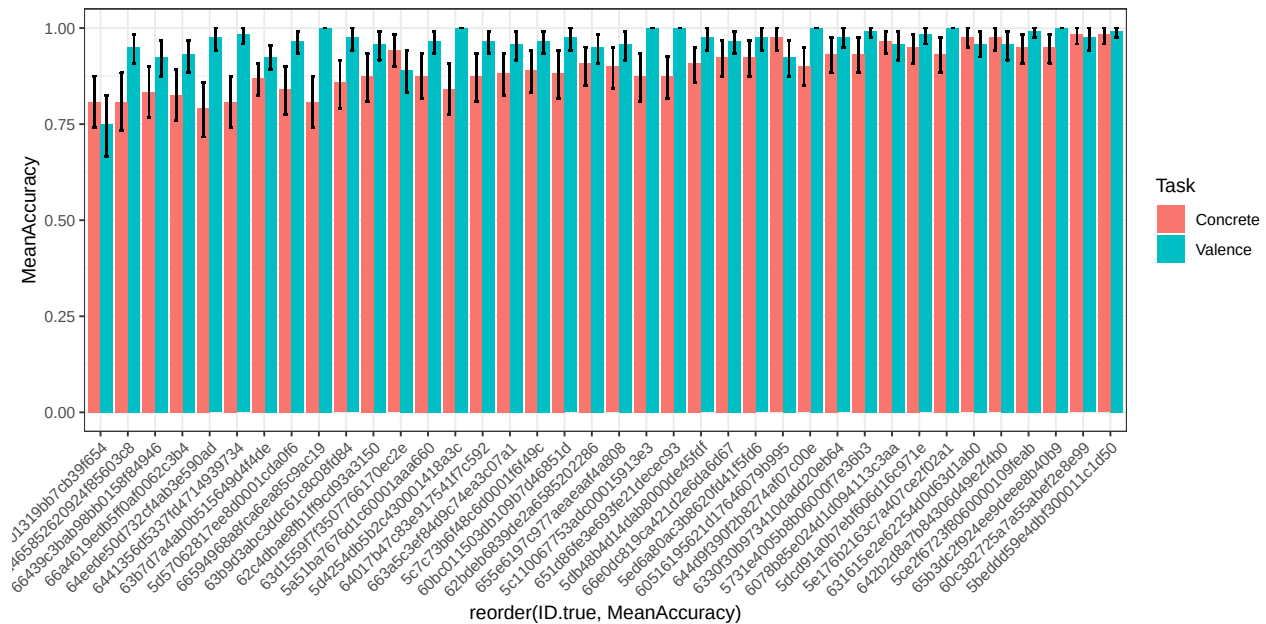


```
## [1] 38
```

Accuracy by Participant

```
agr <- d.inaccurate.removed %>%
  # filter(PennElementType == "Selector") %>%
  select(ID.true, Task, Accuracy) %>%
  group_by(ID.true, Task) %>%
  mutate(MeanAccuracy = mean(Accuracy),
         CILow = ci.low(Accuracy),
         CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
         YMax = MeanAccuracy + CIHigh)

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=reorder(ID.true, MeanAccuracy), y=MeanAccuracy, fill=Task)) +
  geom_bar(position=dodge, stat="identity") +
  geom_errorbar(aes(ymin=YMin, ymax=YMax), width=.25, position=position_dodge(0.9)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# guides(fill = "none")
```

Mean Accuracy by Word / Task

Looking at only the first block

```
agr <- d.inaccurate.removed %>%
  group_by(Task, Word, BlockOrder) %>%
  # filter((Task == "Valence") & (BlockOrder == "VC") |
  #       (Task == "Concrete") & (BlockOrder == "CV"))) %>%
  mutate(MeanAccuracy = mean(Accuracy),
         CILow = ci.low(Accuracy),
         CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
```

```

YMax = MeanAccuracy + CIHigh)

agrr <- agr %>%
  group_by(Word,Task) %>%
  select(Word,Task,MeanAccuracy) %>%
  unique()

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Task,y=MeanAccuracy,fill=BlockOrder)) +
  geom_bar(position=dodge,stat="identity") +
  facet_wrap(~Word,ncol=10) +
  geom_errorbar(aes(ymin=YMin,ymax=YMax),width=.25,position=position_dodge(0.9)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

